Asmi Jain | Software Engineer

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Professional Summary

Software Engineer with experience building enterprise Windows desktop applications and AI-powered web platforms. Delivered production features using C#, .NET Framework, WPF, and MVVM, optimized document-processing performance by 85%, and improved application reliability through asynchronous programming, concurrency management, and production debugging. Experienced with React, FastAPI, PostgreSQL, LangChain, and Groq LLM. Smart India Hackathon 2022 National Winner.

Work Experience

Software Engineer | Celstream Technologies, Bengaluru Aug 2024 – May 2026

- Designed and developed production features for enterprise **Windows desktop applications** using **C#, .NET Framework, WPF, MVVM and XAML** applying Object-Oriented Analysis and Design **(OOAD)** and **SOLID** principles.
- Optimized document-processing performance by **85%** using **asynchronous programming,** Task Parallel Library **(TPL), multithreading** and data binding, significantly improving scalability and UI responsiveness.
- Implemented **lock-file management**, **retry mechanisms**, and **state synchronization** to prevent concurrent document access and ensure data integrity in a **multi-user environment**.
- Developed reusable WPF UI components and delivered **production features** across Document Processing and Batch Manager modules, optimizing **dynamic DataGrid** rendering for **1,000+ records**.
- Investigated production defects through debugging, log analysis, and code tracing; identified root causes, implemented fixes, and validated changes through **regression testing** to ensure stable releases.
- Maintained and enhanced legacy C++/MFC desktop modules by implementing workflow improvements while ensuring backward compatibility and application stability.

Project Trainee | Celstream Technologies, Bengaluru Feb 2024 – Aug 2024

- Contributed to a 100K+ line enterprise **C#/WPF/.NET** codebase and implemented features, **debugging production issues**, and enhancing existing functionality.
- Worked with **WPF, MVVM, Perforce, Visual Studio**, debugging large codebases, and **Agile** software development practices while contributing to enterprise desktop applications.

Technical Skills

- **Languages:** C#, Java, C++, Python, JavaScript, SQL
- **Frameworks & Backend:** .NET Framework, Spring Boot, FastAPI, JWT, SQLAlchemy, REST APIs
- **Frontend:** React.js, HTML5, CSS3
- **Databases:** PostgreSQL, Firebase Realtime Database
- **Core CS:** Data Structures and Algorithms, Object-Oriented Programming, System Design, OOAD, SOLID Principles, Software Engineering
- **AI/ML & Computer Vision:** LangChain, Groq API, TensorFlow, OpenCV, YOLOv3
- **Tools & Cloud:** Git, Perforce, Visual Studio, Azure DevOps, Agile/Scrum

Projects

Make Your Own Adventure Story AI link | GitHub

React, FastAPI, JWT, LangChain, PostgreSQL, SQLAlchemy, Groq, Vercel, Render

- Designed and developed an AI-powered storytelling platform using **React, FastAPI, LangChain, and Groq LLM**, enabling dynamic, context-aware user interactions through structured AI workflows**.**
- Designed scalable **REST API** using FastAPI, SQLAlchemy, PostgreSQL, and **JWT authentication** for secure user management and story generation.
- Improved backend reliability through modular service architecture, **Pydantic** validation, exception handling, and structured request processing.
- Built a responsive **React** application supporting **authenticated** user workflows and dynamic AI story rendering with backend **API** integration for a seamless user experience.

Moving Vehicle Registration Plate Detection SIH 2022 – National Winner

Python, YOLOv3, TensorFlow, OpenCV, Tesseract OCR, Firebase Realtime Database

Smart India Hackathon **2022 – National Winner**, Ministry of Coal, Govt. of India

- Engineered a computer vision pipeline using **YOLOv3, TensorFlow, OpenCV**, and **OCR** to automate vehicle number plate recognition under varying environmental conditions.
- Built **Firebase** real-time **Database** tracking and analytics for monitoring vehicle movement and operational efficiency.
- Optimized **object detection** performance through preprocessing of datasets and custom YOLOv3 model **finetuning** for challenging real-world scenarios

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Education

BNM Institute of Technology, Bangalore 2020 – 2024

- B.E. Computer Science & Engineering | CGPA: **8.5**/10

Sophia Girls Secondary School, Bhilwara 2018 – 2020

- Std. XII (Science) | Percentage: **90%**

Achievements

- Hackathon 5.0 Runner-Up — IEEE-BNMIT Student Branch (2021); built a Mental Health App.
- Secretary, Student Council (12th Std.) — led a 12-member team across student governance and events.